Jackfruit Seed Utilization as a Sustainable Food Source for Malnourished Children and Communities

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Abstract

Malnutrition remains a global challenge, with millions of children lacking sufficient nutrition. Jackfruit, widely grown in South and Southeast Asia, is a nutritious fruit whose seeds are mostly discarded. This paper explores the potential of utilizing jackfruit seeds as a sustainable food source. Based on production data and caloric analysis, the seeds could provide energy to millions of children annually with minimal resources required for preparation. This research highlights a practical approach to reduce food waste, improve nutrition, and contribute to local economies.

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1. Introduction

Despite advancements in agriculture, 50–100 million people worldwide still experience food scarcity, with children disproportionately affected. Bangladesh and India are among the largest producers of jackfruit, yet a significant portion of the fruit is wasted, particularly the seeds.

Jackfruit seeds are nutritionally dense, easy to process, and widely available, making them a promising resource to address malnutrition in resource-limited regions. This paper presents calculations and an implementation model for converting discarded jackfruit seeds into a practical food solution.

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2. Problem Statement

- Millions of children suffer from malnutrition worldwide.
- Conventional food aid requires water, oil, and infrastructure, which may be limited in crisis zones.
- Jackfruit seeds, representing ~10–12% of total fruit weight, are discarded as waste, representing lost nutritional and economic potential.

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3. Jackfruit Production and Seed Availability

- **Annual Global Production:** 422 million jackfruits
- **Average Weight per Fruit:** 8–10 kg

- **Seed Weight:** ~1 kg per fruit
- **Major Producers:** Bangladesh, India, Indonesia, Sri Lanka

Assuming 80% of seeds are collected:

- Total seeds: 337.6 million kg
- Total energy content: ~640 billion kilocalories (kcal)

This energy could provide:

- 1,200 kcal per day to 533.7 million children, or
- A full year of nutrition for ~1.4 million children

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4. Nutritional Analysis of Jackfruit Seeds (per 100 g raw seeds)

- Energy: 190 kcal
- Protein: 7–9 g
- Carbohydrate: 35–40 g
- Fiber: 1.5–2 g
- Fat: 0.4 g
- High levels of zinc, iron, and other vitamins

Preparation: Roasting or baking requires no water or oil, making it suitable for regions with scarce resources.

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5. Proposed Implementation Model

1. **Seed Collection:** Collect seeds from fruit processing centers, markets, and farms.
2. **Processing:** Clean, dry, and roast seeds for consumption.
3. **Distribution:** Supply to orphanages, schools, refugee camps, and food aid programs.
4. **Scale-up:** Increase jackfruit cultivation in Bangladesh and India to expand production.

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6. Economic and Livestock Benefits

- Exporting jackfruit juice or processed seeds could strengthen local economies.
- Residual fruit waste can feed livestock, supporting milk and meat production.

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7. Impact and Sustainability

- Provides a low-cost, nutrient-rich food source.
- Reduces food waste.
- Requires minimal water, oil, and infrastructure.
- Supports both human nutrition and livestock management.

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8. Limitations and Future Work

- Infrastructure for collection, processing, and distribution is required.
- Logistics in conflict zones or disaster-affected areas may be challenging.
- Future research could develop shelf-stable seed products, fortified food items, and pilot large-scale programs.

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9. Conclusion

Jackfruit seed utilization represents a **practical and sustainable solution** to address malnutrition, particularly in developing countries. By converting a widely discarded resource into a nutrient-rich food, this approach has the potential to feed millions of children while contributing to local economies and agriculture. Even partial implementation could have a significant humanitarian impact.

Author's Note

This research is an original work by the author and has not been published elsewhere.